MTH601 Operations Research

Question No : 1 of 52

The cost matrix in assignment problem is always diagonal matrix

Created By Mahar Afaq Safdar Muhammadi

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Answer (Please select your correct option)

- square matrix — correct
- identity matrix
- zero matrix

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MTH601 Operations Research

Question No : 2 of 52

Which of the following binary operation in assignment problem among all the elements in the given profit matrix from the highest element in the matrix

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Answer (Please select your correct option)

- Subtraction **correct**
- Division
- Multiplication
- Addition

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Question No : 3 of 52

During a replacement if the value of money decreases at the rate of **3**% then the present worth factor of unit amount to be spent after one year is given by

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Answer (Please select your correct option)

- 0.25
- 0.333
- 0.9708 **correct**
- 4

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

In sequencing problems, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machines problems

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Answer (Please select your correct option)

- $3n$ correct
- $3n!$
- $2 \times 3 = 6$
- 2

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MTH601 Operations Research

Question No : 5 of 52

If the money carries a rate of interest of **12%** per year, the present worth factor of one rupee due in one year is

VuA

Answer (Please select your correct option)

- 0.08333
- 0.89285 **correct**
- 0.0769
- 13

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Question No : 6 of 52

Degeneracy in a 5×6 transportation problem occurs when the number of occupied cell is less than

VuA

Answer (Please select your correct option)

- 20 but greater than 10
- 10 **correct**
- Zero
- Infinity

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Question No : 7 of 52

We go in probabilistic replacement model when period between installation and failure is

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Answer (Please select your correct option)

- varying exponentially
- varying linearly — correct
- Constant
- is not constant (varying arbitrarily)

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Question No : 8 of 52

Any set of non negative allocations ($X_{ij} > 0$) which satisfies the row and column sum is called a _____ solution.

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Answer (Please select your correct option)

- feasible
- non basic feasible
- basic infeasible
- optimal

correct

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Question No : 9 of 52 Marks: 1 (Budgeted

If a basic feasible solution contains less than "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

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Answer (Please select your correct option)

- Degenerate **correct**
- Multiple Solutions
- Non degenerate
- Alternative Optima

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MTH601 Operations Research

Question No : 10 of 52

Consider the following cost table:

	Destinations				
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10

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Answer (Please select your correct option)

- 10
- 15
- 25
- 5

correct

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Question No : 11 of 52

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10

VuA

Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

correct

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Question No : 11 of 52

	S3	3 10	3	5	15
	Demand	10	20	20	

Using Vogel's Approximation Method:
These below are the three cells which can be next cell:

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Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

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Question No : 11 of 52

I. S3 → D2
II. S2 → D2
III. S1 → D3

Which one is the correct option?

VuA

Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

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Question No : 12 of 52

We go in probabilistic replacement model when period between installation and failure is

VuA

Answer (Please select your correct option)

- varying exponentially
- varying linearly

correct

- Constant
- is not constant (varying arbitrarily)

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Question No : 13 of 52

If the mean arrival and mean service rates are **4** and **7** respectively in a queue then expected waiting time in the system is

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Answer (Please select your correct option)

- $\frac{1}{3}$ correct
- 3
- 28
- $\frac{7}{4}$

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

In a bank, every 15 minutes one customer arrives for cashing the cheque. The staff in the payment counter takes only 10 minutes for serving a customer on an average, then the service rate " μ" = _____.

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Answer (Please select your correct option)

- 6 per hour — correct
- 4 per hour
- 10 per hour
- 1/6 per hour

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

A duplicating machine maintained for office use is used and operated by people in the office who need to make copies. Since the work to be copied varies in length (number of pages of the original) and copies required, the service rate is randomly distributed, the arrival rate is 5 per hour and the service rate is 10 per hour then the equipment utilization " ρ " is equal to

VuAnswers.com

Answer (Please select your correct option)

- 0.50 **correct**
- 0.20
- 5
- 2

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the average rate " λ " is

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Answer (Please select your correct option)

- 1/8 = 0.125 **correct**
- 8
- 4
- ¼ = 0.25

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the mean service time " μ " is

VuAnswers.com

Answer (Please select your correct option)

- ½ = 0.5 **correct**
- 4
- ¼ = 0.25
- 2

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MTH601 Operations Research

Question No : 18 of 52

If "N_i" be the Number of replacement made at the end of the ith week and "P_j" be the probability of failure during the ith week, then N_1 = -------

VuA

Answer (Please select your correct option)

- N_0P_1 **correct**
- N_1P_1
- N_0P_0
- N_1P_2

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Question No : 19 of 52

The present worth of a rupee to be spent after a year is denoted by v and given by

VuA

Answer (Please select your correct option)

- $v = 1/(1+r)$ **correct**
- $v = (1+r)$
- $v = (1+r)/10$
- $v = (1/r)$

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Question No : 20 of 52

Formula for a geometric series "$x + vx + v^2x + \ldots + v^{n-1}x$" is

VuA

Answer (Please select your correct option)

- $P(n) = x \dfrac{1 - v^n}{1 - v}$ — correct
- $P(n) = \dfrac{1 - v^n}{1 - v}$
- $P(n) = x \dfrac{1 - v}{1 - v^n}$
- $P(n) = \dfrac{1 - v}{1 - v^n}$

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Question No : 21 of 52

Product of 'item cost' and 'ordered item' is

Answer (Please select your correct option)

- Crash cost
- Cost period

correct

- Set up cost
- Shortage cost

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MTH601 Operations Research

Question No : 22 of 52

Standard Deviation (S.D.) = _____ MAD

VuA

Answer (Please select your correct option)

- $\sqrt{\frac{2\pi}{3}}$
- $\sqrt{\frac{2}{\pi}}$
- $\sqrt{\frac{\pi}{2}}$ correct
- $\sqrt{\frac{3\pi}{2}}$

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Question No : 23 of 52

MAD = --------- S.D.

VuA

Answer (Please select your correct option)

- $\sqrt{\frac{2}{\pi}}$ **correct**
- $\sqrt{\frac{\pi}{2}}$
- $\sqrt{\frac{2\pi}{3}}$
- $\sqrt{\frac{3\pi}{2}}$

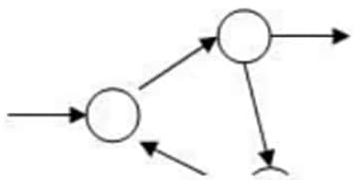
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Question No : 24 of 52

The following network is an example of



Answer (Please select your correct option)

- Redundancy
- Dangling **correct**
- Cycling
- Dummy

VuA

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Question No : 25 of 52

For any activity backward pass computations provide its

VuA

Answer (Please select your correct option)

- Earliest start times
- Latest start times

correct

- Moderate start times
- Completion time

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Question No : 26 of 52

Best possible time estimate that a given activity would take under normal conditions which often exist, is called

VuA

Answer (Please select your correct option)

- Most Likely time estimate
- Pessimistic time estimate correct
- Smallest time estimate
- Activity time estimate

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Question No : 27 of 52

In PERT, activity time estimates are distributed according to

VuA

Answer (Please select your correct option)

- Beta Distribution — correct
- Normal Distribution
- Poisson distribution
- Binomial Distribution

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Question No : 28 of 52

Solution region for the constraint $y \leq 0$ is the

VuA

Answer (Please select your correct option)

- Half plane below y-axis
- Half plane below the line: $y = 0$
- Set of all those points where ordinates are non-positive
- All are equivalent

correct

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Question No : 29 of 52

In a linear Programming Problem (LPP), which of the following must be hold?

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Answer (Please select your correct option)

- Only objective function is linear
- Both objective function and constraints are linear **correct**
- Only constraints needs to be linear
- At least one of objective function or constraint should be linear

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Question No : 30 of 52 Marks: 1 (Budgeted Tin

If a company manufacture 'x' units of product 'A' and 'y' units of 'B' with associated profits of Rs.5 and Rs.3 then which of the following is the objective function to maximize is the profit?

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Answer (Please select your correct option)

- $z = 15xy$
- $z = 5x - 3y$
- $z = 3x - 5y$
- $z = 5x + 3y$

correct

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Question No : 31 of 52

The _____ variable is chosen by examining the cost coefficents in the objective function.

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Answer (Please select your correct option)

- entering — correct
- leaving
- positive slack
- negative slack

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Question No : 32 of 52

While solving a linear programming problem by big M – Method, traditionally the _____ variables are chosen in the initial basic feasible solution.

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Answer (Please select your correct option)

- neagtive slack
- positive slack
- entering
- artificial correct

Subjective

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Question No : 33 of 52 Marks: 1 (Budgeted Ti

In the big-M method, if the introduced _____ variables do not leave the basis in the final iteration, then this indicates that the give linear programming problem cant be optimized.

VuAnswers.com

Answer (Please select your correct option)

- entering
- positive slack
- negative slack
- artificial

correct

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Question No : 34 of 52

Zero valued artificial variables may appear as _____ variables in the final solution, when one or more of the original constraints equations is redundant.

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Answer (Please select your correct option)

- non basic
- basic

correct

- slacks
- surplus
- artificial

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

By using Two phase method to solve a linear programming problem, in phase I, a new objective function is formed by assigning on left hand side, zero to every original variable (including slack and surplus variables) and -------- to each of the artificial variables.

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Answer (Please select your correct option)

- M
- −M

correct

- +1
- −1

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Question No : 36 of 52

For the linear programming problem;

$\text{Max}Z = 2x + 3y$

Subject to

$$\left.\begin{array}{l} x \geq 2 \\ y \leq 3 \\ x, y \geq 0 \end{array}\right\} \Rightarrow \left.\begin{array}{l} x - s_1 + A = 2 \\ y + s_2 = 3 \\ x, y, s1, s2, A \geq 0 \end{array}\right\}$$

VuA

Answer (Please select your correct option)

- $\text{Max}Z = 2x + 3y + A$
- $\text{Min}Z = 2x + 3y + A$
- $\text{Max}Z = A$
- $\text{Min}Z = A$

Ma b

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Question No : 36 of 52

$$\left.\begin{matrix} y \le 3 \\ x, y \ge 0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} y + s_2 = 3 \\ x, y, s1, s2, A \ge 0 \end{matrix}\right\}$$

Which of the following is associated objective function of the 1st phase?

Answer (Please select your correct option)

- $\text{Max}Z = 2x + 3y + A$
- $\text{Min}Z = 2x + 3y + A$
- $\text{Max}Z = A$
- $\text{Min}Z = A$

correct

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Question No : 37 of 52

The inequality $2x + 3y \geq 18$ is equivalent to

Answer (Please select your correct option)

- $2x + 3y \leq -18$
- $-2x - 3y \geq 18$
- $-2x - 3y \geq -18$
- $-2x - 3y \leq -18$ correct

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Question No : 38 of 52

In which of the following models, Simplex algorithm is not preferred to use due to laborious computations?

Answer (Please select your correct option)

- Transportations models
- Degenerate Linear models

correct

- Non-degenerate Linear models
- Dual or unbounded linear models

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Question No : 39 of 52

Transportations models consist of ------- like the production centers and --------- which may be the sales centers.

Answer (Please select your correct option)

- (sinks, sources)
- (sources, sinks) **correct**
- (origins, sources)
- (sinks, destinations)

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Question No : 40 of 52

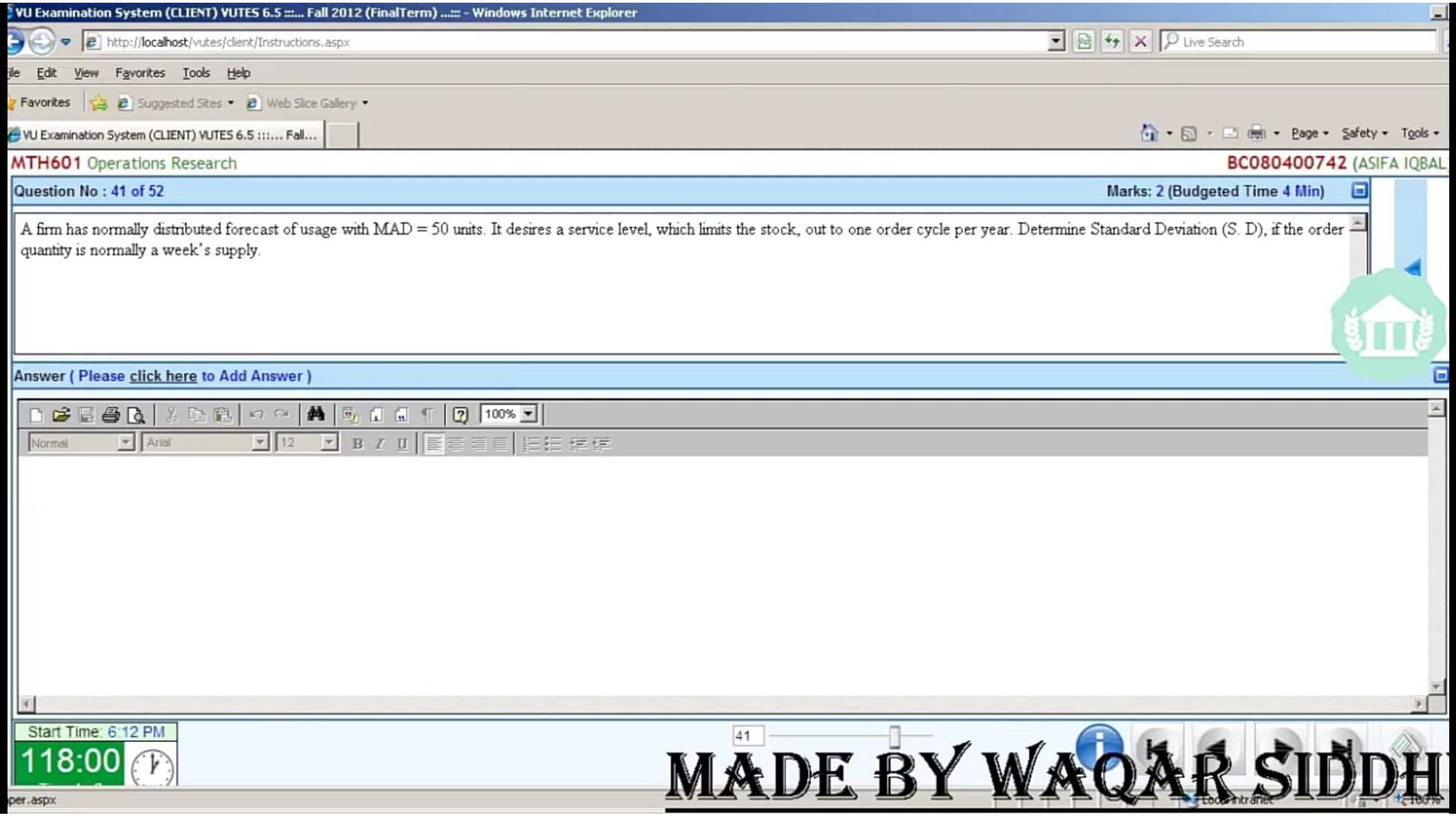
To convert the transportation problem into a maximization model we have to----------.

Answer (Please select your correct option)

- write the inverse of the coefficient matrix
- multiply the feasibility condition by −1
- multiply the coefficient matrix by −1
- We can't convert the transportation problem into a maximization problem, as it is basically a minimization prob

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

A company centre has got four expert programmers. The centre needs four application programmes to be developed. The head of the computer centre, after studying carefully the programme's to be developed, estimate the computer time in minutes required by the respective experts to develop the application programmes as follows.

Programmers	A	B	C	D

Answer (Please click here to Add Answer)

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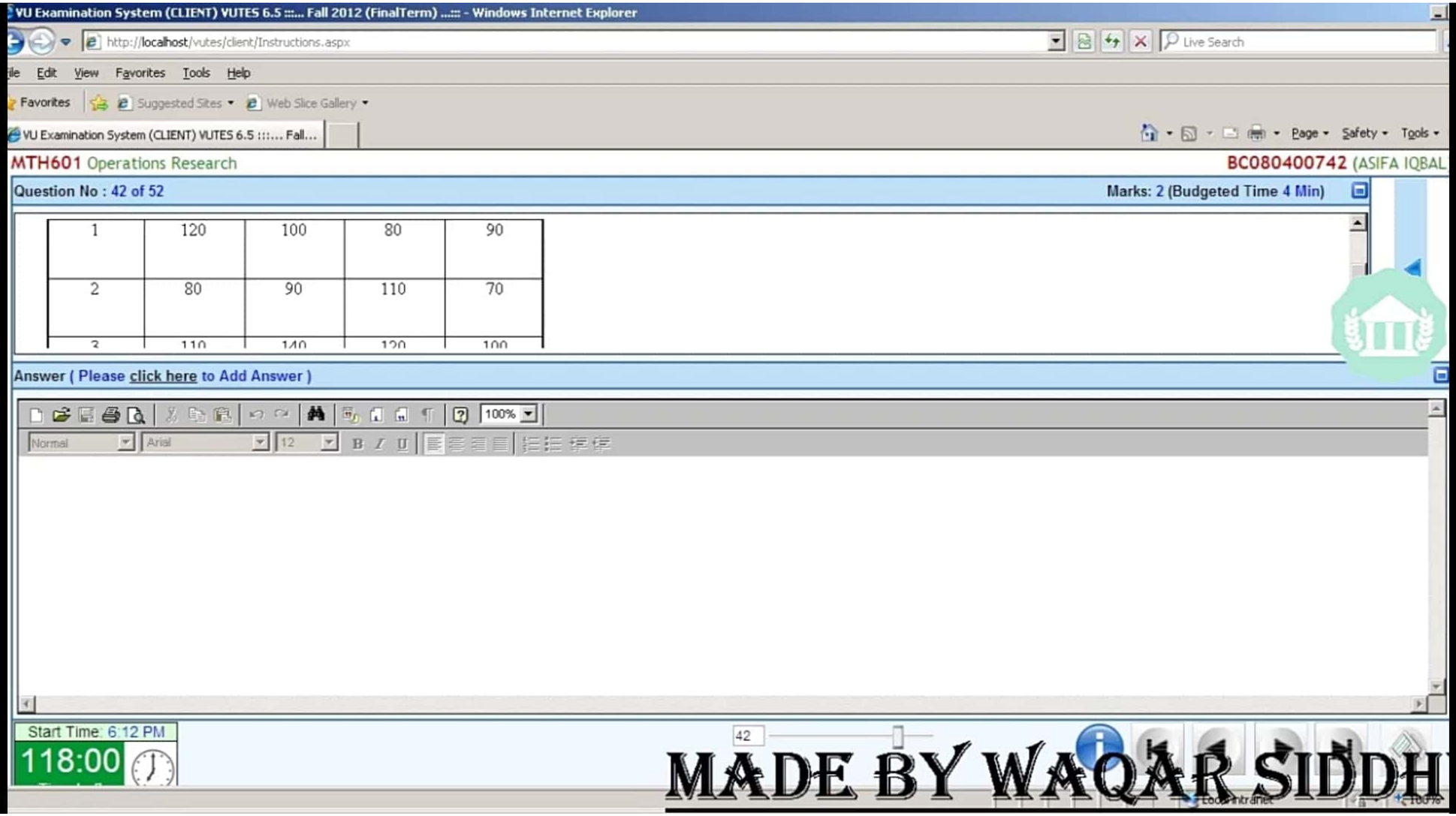
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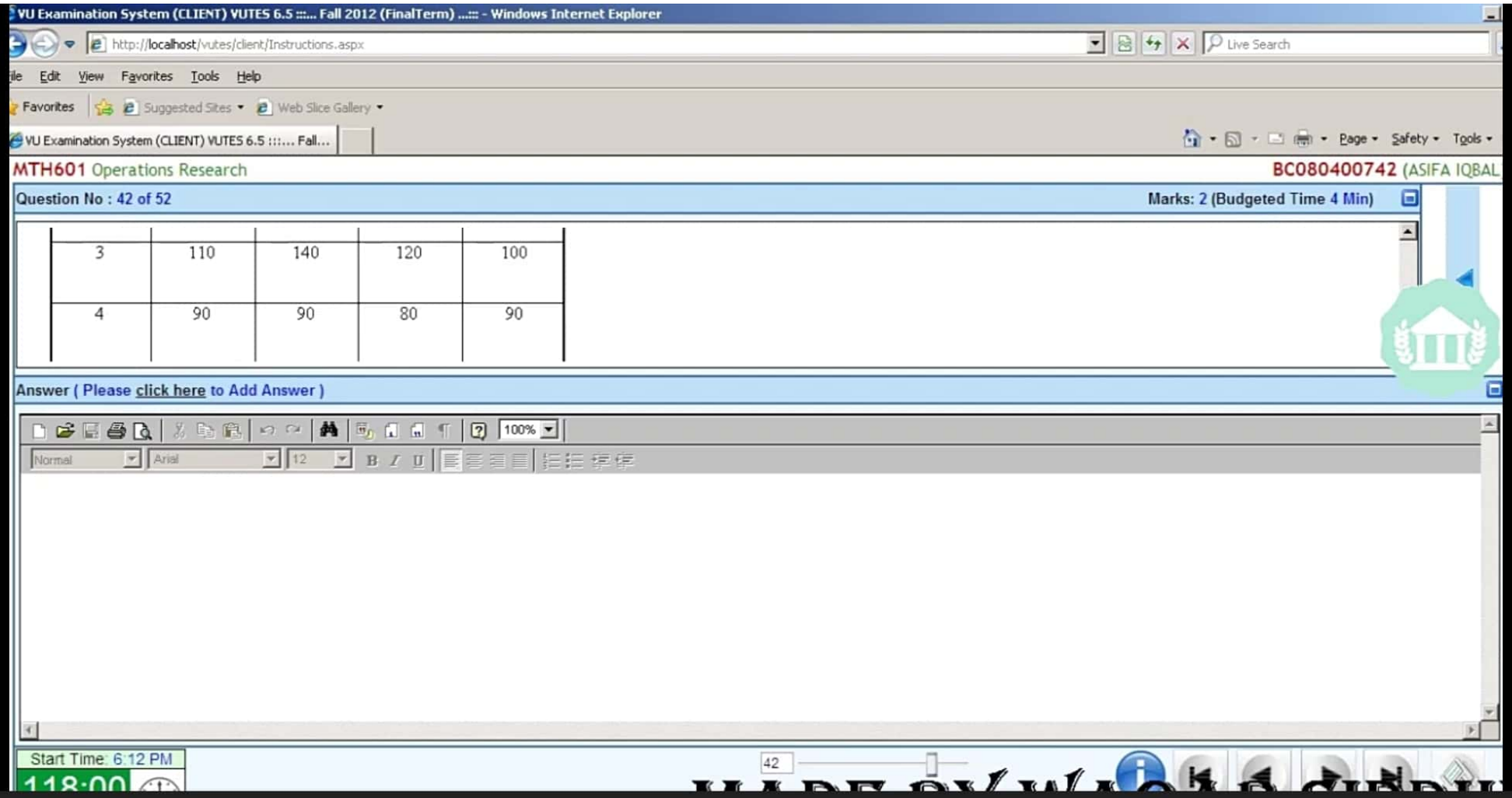
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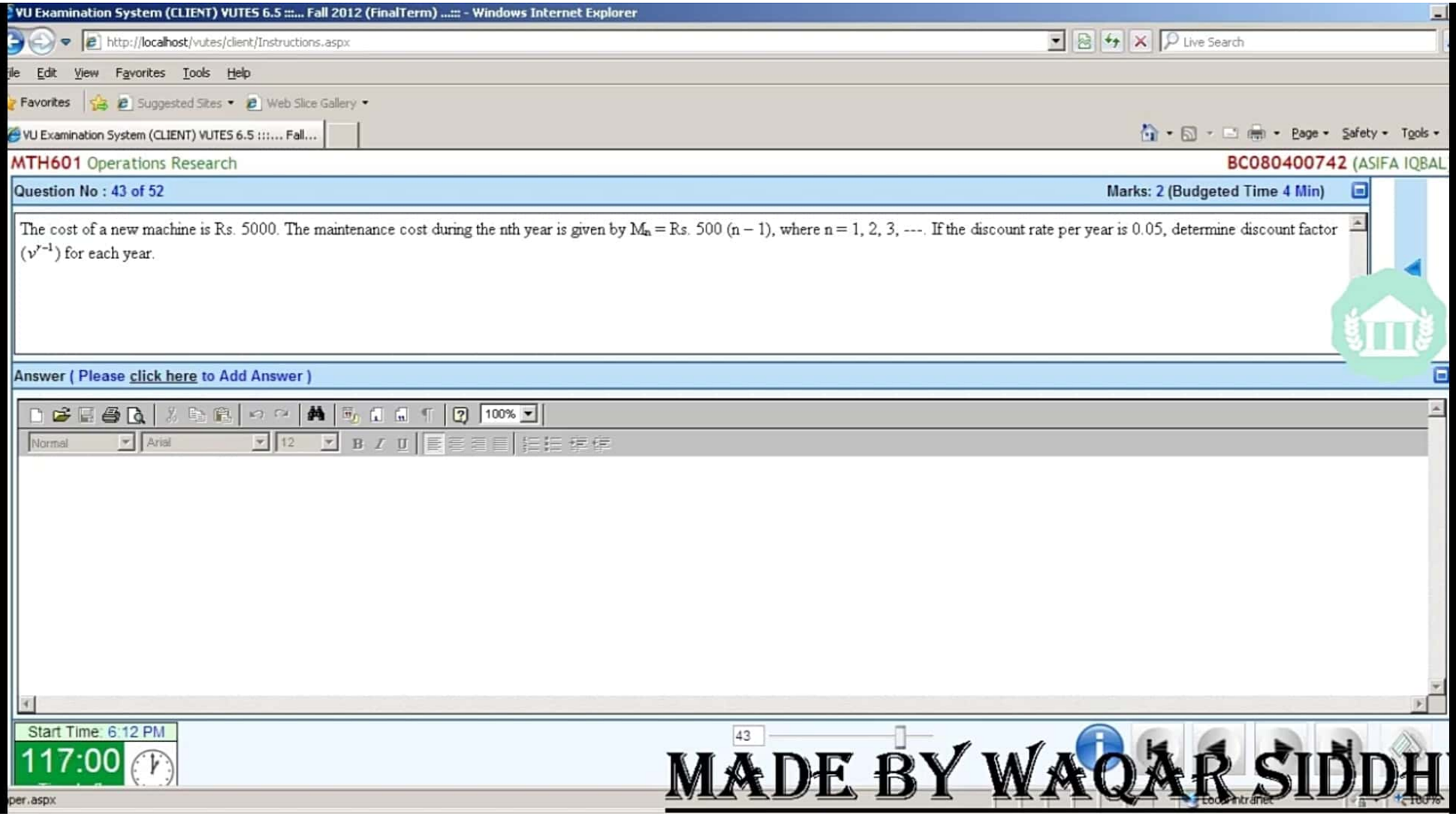


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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Determine whether the following Transportation model has initial feasible solution?

	D_1	D_2	D_3	D_4	Supply
O_1	x_{11}	x_{12}	x_{13}	x_{14}	6

Answer (Please click here to Add Answer)

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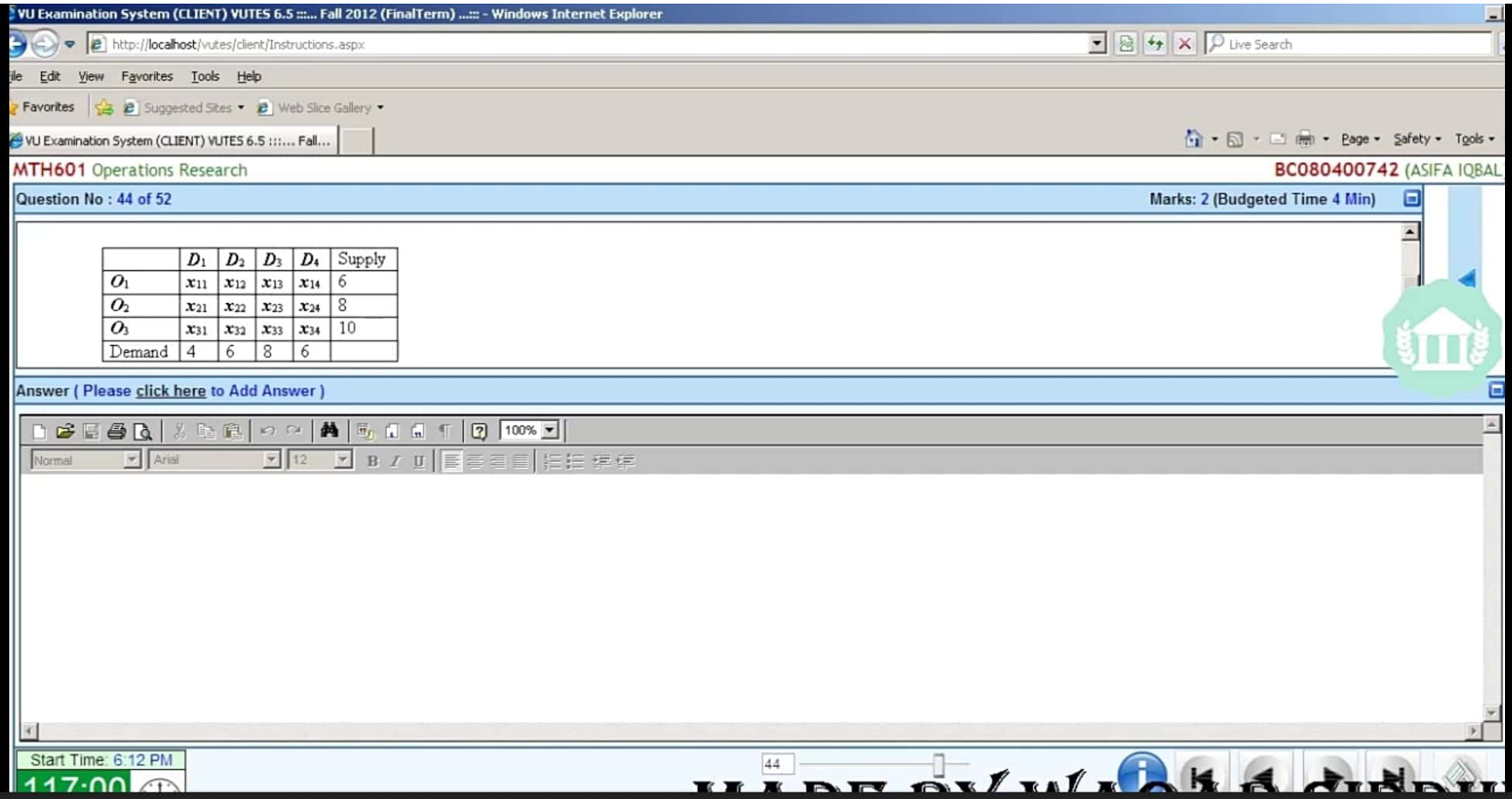
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	D_1	D_2	D_3	D_4	Supply
O_1	x_{11}	x_{12}	x_{13}	x_{14}	6
O_2	x_{21}	x_{22}	x_{23}	x_{24}	8
O_3	x_{31}	x_{32}	x_{33}	x_{34}	10
Demand	4	6	8	6	

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Salesman	Region					
	1	2	3	4	5	6
A	5	0	25	0	15	0
B	**0**	50	0	25	30	45
C	35	15	55	**0**	40	75
D	10	30	5	30	15	**0**

Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

C	35	15	55	**0**	40	75
D	10	30	5	30	15	**0**
E	25	**0**	30	55	20	30
F	35	35	20	10	**0**	55

Do next step by applying Hungarian method.

Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

An oil company has 8 unit of money available for exploration of three sites. If oil is present at a site, the probability of finding it depends upon the amount allocated for exploiting the site as given below:

	0	1	2	3	4	5	6	7	8
Site I	0.0	0.0	0.1	0.2	0.3	0.5	0.7	0.9	1.0
Site II	0.0	0.1	0.2	0.3	0.4	0.6	0.7	0.8	1.0

Answer (Please click here to Add Answer)

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MTH601 Operations Research BC080400742 (ASIFA IQBAL

Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Site I	0.0	0.0	0.1	0.2	0.3	0.5	0.7	0.9	1.0
Site II	0.0	0.1	0.2	0.3	0.4	0.6	0.7	0.8	1.0
Site III	0.0	0.1	0.1	0.2	0.3	0.5	0.8	0.9	1.0

Answer (Please click here to Add Answer)

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MTH601 Operations Research BC080400742 (ASIFA IQBAL

Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

The probability that the oil exits at sites I, II and III is 0.4, 0.3 and 0.2 respectively, we have to find the optimal allocating of money. Stage I is given below; **only do stage II.**

Stage I

Max. $Z = 0.4P_1(x_1) + 0.3\ P_2(x_2)$

Subject to:

$$x_1 + x_2 + x_3 \leq 8$$

Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

No. of boxes x_1	0	1	2	3	4	5	6	7	8
$f_1(x_1)$	0	0	4	8	12	20	28	36	40

Answer (Please click here to Add Answer)

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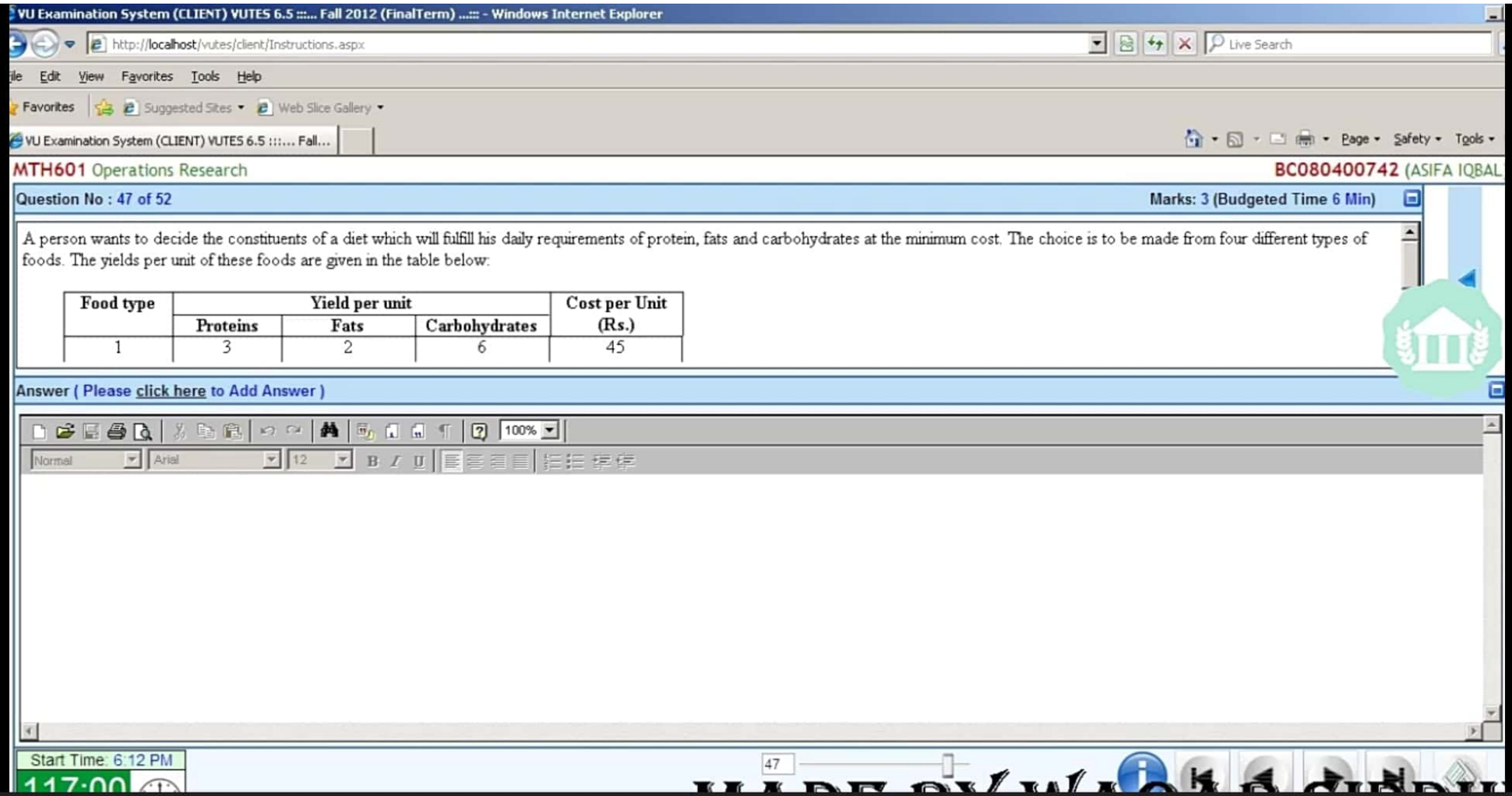
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Operations Research (MTH601)

Question: 1 (Marks: 1) Attempted Questions: 0 Total Questions: 52

Under the individual replacement policy an item is replaced on its

Choices:

- Success
- Failure
- Maximum utilization
- Minimum utilization

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Operations Research (MTH601)

Question: 2 (Marks: 1) Attempted Questions: 1 Total Questions: 52

During a replacement if the value of money decreases at the rate of 3% then the present worth factor of unit amount to be spent after one year is given by

Choices:

0.25

0.333

0.9708

4

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Operations Research (MTH601)

Question: 3 (Marks: 1) Attempted Questions: 2 Total Questions: 52

In sequencing problems, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machines problems

Choices:

$3n$

$3n!$

$2 \times 3 = 6$

2

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Operations Research (MTH601)

Question: 4 (Marks: 1) Attempted Questions: 3 Total Questions: 52

The name of the method in getting the optimum assignment is

Choices:

Simplex method

North west corner method

Vogal approximation method

Hungarian method

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Operations Research (MTH601)

Question: 5 (Marks: 1) Attempted Questions: 4 Total Questions: 52

If the mean arrival and mean service rates are 4 and 7 respectively in a queue then expected waiting time in the system is

Choices:

1/3

28

3

7/4

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Operations Research (MTH601)

Question: 6 (Marks: 1) Attempted Questions: 5 Total Questions: 52

The representation of reality in some physical form or in some form of mathematical equations may be discussed under the topic entitled as

Choices:

- Transportation problems
- Game theory
- Simulation
- Replacement problems

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Operations Research (MTH601)

Question: 7 (Marks: 1) Attempted Questions: 5 Total Questions: 52

Transportation technique or the simplex method cannot be used to solve the assignment problem because of

Choices:

Degeneracy

Involvement of dummy activities

Looping among the entries of assignment matrix

Inequilibrium between demand and supply

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Operations Research (MTH601)

Question: 8 (Marks: 1) Attempted Questions: 6 Total Questions: 52

Search and cutting methods are used to solve following types of problems

Choices:

Transportation

Assignment

Integer Programming

Queuing problems

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Operations Research (MTH601)

Question: 9 (Marks: 1) Attempted Questions: 7 Total Questions: 52

CPM can help you figure out EXCEPT:

Choices:

how long your complex project will take to complete

which activities are "critical," meaning that they have to be done on time or else the whole project will take longer

what is the most costly way to speed up the project

you can try to speed up the project

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Operations Research (MTH601)

Question: 11 (Marks: 1) Attempted Questions: 9 Total Questions: 52

If a basic feasible solution contains less than "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

Choices:

- Degenerate
- Multiple Solutions
- Non degenerate
- Alternative Optima

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Operations Research (MTH601)

Question: 10 (Marks: 1) Attempted Questions: 8 Total Questions: 52

The Vogel Approximation method is an iterative procedure for computing a _____ solution of the transportation problem.

Choices:

basic feasible

non basic feasible

basic infeasible

optimal

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Operations Research (MTH601)

Question: 12 (Marks: 1) Attempted Questions: 10 Total Questions: 52

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10
	S3	3	3	5	15
	Demand	10	20	20	

Choices:

10

15

25

5

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Operations Research (MTH601)

Question: 14 (Marks: 1) Attempted Questions: 11 Total Questions: 52

The name of the method in getting the optimum assignment is

Choices:

Simplex method

North - West corner method

Vogel's approximation method

Hungarian method

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Operations Research (MTH601)

Question: 15 (Marks: 1) Attempted Questions: 12 Total Questions: 52

Which of the following binary operation in assignment problem among all the elements in the given profit matrix from the highest element in the matrix

Choices:

- Subtraction
- Division
- Multiplication
- Addition

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Operations Research (MTH601)

Question: 17 (Marks: 1) Attempted Questions: 14 Total Questions: 52

In assignment problem, each job requires exactly one resource.

Choices:

True

False

May not be true

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Operations Research (MTH601)

Question: 18 (Marks: 1) Attempted Questions: 14 Total Questions: 52

If "N_i" be the Number of replacement made at the end of the ith week and "P_j" be the probability of failure during the ith week, then N_2 = ----------.

Choices:

- $N_0P_2 + N_1P_1$
- $N_0P_1 + N_1P_2$
- $N_0P_2 + N_1P_2$
- $N_1P_2 + N_1P_1$

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Operations Research (MTH601)

Question: 19 (Marks: 1) Attempted Questions: 15 Total Questions: 52

In replacement Models, Capital Cost = (Item / Machine Cots) – (----------).

Choices:

- Resale Cost
- Running Cost
- Maintenance Cost
- Operating Cost

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Operations Research (MTH601)

Question: 22 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Let FS = Free Slack, TS = Total Slack, INDS = Independent Slack, we have the relations

I. TS ≤ FS
II. INDS ≤ FS
III. FS ≤ TS

Then which relation is true

Choices:

- Only I
- Only II
- Only III
- Only I & II
- Only II & III

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Operations Research (MTH601)

Question: 23 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Which of the following give the excess of available time over the activity time when all jobs start as early as possible?

Choices:

- Total float
- Free float
- Independent float
- Early finish times

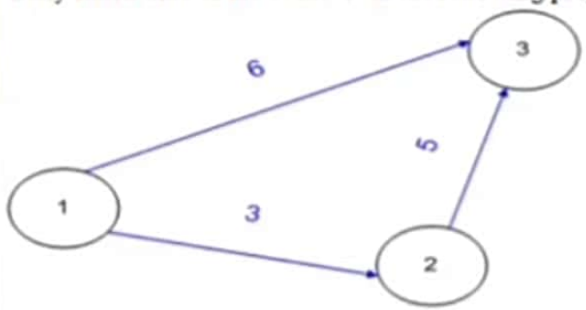
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Operations Research (MTH601)

Question: 24 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Early finish time of the event '3' in the following project is -------.



Choices:

14 days

2 days

8 days

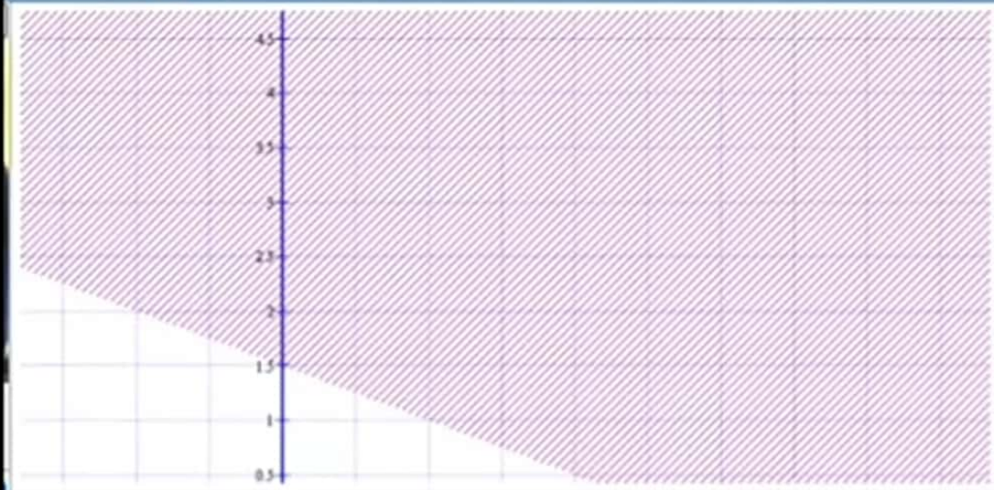
6 days

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Operations Research (MTH601)

Question: 25 (Marks: 1) Attempted Questions: 16 Total Questions: 52



Choices:

$x+2y \leq 3$

$x+2y \geq 3$

$x-2y \leq 3$

$x-2y \geq 3$

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Operations Research (MTH601)

Question: 26 (Marks: 1) Attempted Questions: 16 Total Questions: 52

It is a property of simplex method that there always exist infinite numbers of basic feasible solutions.

Choices:

True

False

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Operations Research (MTH601)

Question: 27 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If the objective function of a linear programming problem needs further improvement then which of the following will have to proceed?

Choices:

a new decision variable to be entered and other decision variable to leave the basis

a new slack variable to be entered and other slack to leave the basis

a new non-degenerate variable to be entered and other non-generate to leave the basis

a new basic variable to be included and other basic variable t

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Operations Research (MTH601)

Question: 28 (Marks: 1) Attempted Questions: 16 Total Questions: 52

The constrained $3x_1 + 2x_2 \leq 18$ in standard form can be written as:
(Where R is artificial variable and S_1 is Slack variable)

Choices:

$3x_1 + 2x_2 + S_1 + R = 18$

$3x_1 + 2x_2 + S_1 - R = 18$

$3x_1 + 2x_2 - S_1 + R = 18$

$3x_1 + 2x_2 + S_1 = 18$

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Operations Research (MTH601)

Question: 29 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If no _____ variable appears in the basis and the optimality conditions are satisfied, then the current solution will be optimal one.

Choices:

artificial

non basic

slack

surplus

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Operations Research (MTH601)

Question: 30 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Min. Z = 0 and no artificial variable appears in the basic variables. _____ solution to original problem has been found and we proceed to phase II method.

Choices:

A basic feasible

An optimal

An infeasible

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Operations Research (MTH601)

Question: 31 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If the problem has a _____ solution such that the minimum value of the objective function is zero along with zero values of artificial variables. Then proceed to phase II method.

Choices:

unbounded

feasible

non feasible

de-generate feasible solution

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Operations Research (MTH601)

Question: 32 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If the degeneracy arises in the initial stage, then which of the following is true?

Choices:

One of the basic variables is zero.

More than one variable is eligible to leave the basis.

Tie in the ratio of each row,by taking right hand side of each row and dividing by the corresponding element of the key column.

Iterative Cycling of basic solution exists without reaching th

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Operations Research (MTH601)

Question: 33 (Marks: 1) Attempted Questions: 16 Total Questions: 52

For a linear programming problem, the Unboundedness is related to which of the following?

Choices:

Solution space only

Objective value only

Neither to solution space nor to objective values

Both solution space and objective value

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Operations Research (MTH601)

Question: 34 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Consider the following simplex table corresponding to maximization problem

Basic	x_1	x_2	x_3	S_1	S_2	S_3	R.H.S
Z	-4	-3	-2	0	0	0	0
x_1	0	0	1	1	0	0	8
S_2	-2	1	3	0	1	0	9
S_3	0	2	1	0	0	1	3

Then this is the indication that

Choices:

- Problem has Unbounded Solutions
- Problem has Infeasible Solutions
- Problem has temporarily degenerate
- Problem has Multiple Optimal Solutions

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Operations Research (MTH601)

Question: 35 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Consider the following simplex table corresponding to the maximization problem

Basic	x_1	x_2	x_3	x_4	A_1	A_2	R.H.S
Z	0	0	M + 1/2	M	16M – 11	0	-0.6M – 5.7
x_1	1	0	5	0	-1	0	3
x_2	0	1	-5	0	3	0	9
A_2	0	0	-1	-1	-0.6	1	0.6

This is the indication that problem has----------.

Choices:

infeasible Solution

temporarily Degenerate Solution

unbounded Solution

degenerate Solution

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Operations Research (MTH601)

Question: 36 (Marks: 1) Attempted Questions: 16 Total Questions: 52

In the Simplex table, if the coefficient of non-basic variable says x_1 in the z-row is zero, then it indicates that value of z ----------.

Choices:

- increase
- decreases
- does not change
- becomes infinity

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Operations Research (MTH601)

Question: 37 (Marks: 1) Attempted Questions: 16 Total Questions: 52

A primal's constraint equation produces _____ dual variable.

Choices:

non basic

an unrestricted

artificial

basic

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Operations Research (MTH601)

Question: 38 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Transportation problem is a special class of linear programming problem in which transport a commodity from the source to a destination in such a way that the total transportation cost is -------.

Choices:

Zero

Maximum

Minimum

Average

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Operations Research (MTH601)

Question: 39 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Many decision making problems involve a process that takes place in multiple stages in such a way that at each stage, the process is dependent on the strategy chosen. Such types of problems are called

Choices:

- Linear Programming problems
- Integer programming problems
- Dynamic programming problems
- Assignment problems

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Operations Research (MTH601)

Question: 40 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If shortage cost is infinity, then .....

Choices:

no shortages are allowed

no inventory carrying cost is allowed

Ordering cost is zero

Purchase cost = Carrying cost

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Operations Research (MTH601)

Question: 41 (Marks: 2) Attempted Questions: 16 Total Questions: 52

	1	2	3	Supply
1	0 5	2	1 1	6
2	2	1 5	5 2	7
3	2	4	3 7	7
Demand	5	5	10	

What is the total shipping cost of the above table?

Answer:

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0 5	2	1 1	6
2	1 5	5 2	7
2	4	3 7	7
5	5	10	

To find optimality condition, we use UV Multiplier Process.

Find a) $U_2 + V_2$

b) $U_3 + V_3$

Answer:

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Operations Research (MTH601)

Question: 43 (Marks: 2) Attempted Questions: 16 Total Questions: 52

The cost of a new machine is Rs. 5000. The maintenance cost during the nth year is given by M_n = Rs. 500 (n – 1), where n = 1, 2, 3, ---. If the discount rate per year is 0.05, determine discount factor (v^{t-1}) for each year.

Answer:

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Operations Research (MTH601)

Question: 44 (Marks: 2) Attempted Questions: 16 Total Questions: 52

What are the key issues of inventory model? Also describe its objective.

Answer:

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Programmers	A	B	C	D
1	30	10	0	10
2	0	10	40	0
3	0	30	20	0
4	0	0	0	10

Assign each job to each programmer.

Answer:

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Operations Research (MTH601)

Question: 46 (Marks: 3) Attempted Questions: 16 Total Questions: 52

A company has a machine whose cost is Rs. 30,000. Its maintenance cost and resale value at the end of different years are as given below:

Years.	1	2	3	4	5	6
Maintenance Cost.	4500	4700	5000	5500	6500	7500
Resale Value	27000	25300	24000	21000	18000	13000

Determine capital cost for each year.

Answer:

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Operations Research (MTH601)

Question: 47 (Marks: 3) Attempted Questions: 16 Total Questions: 52

Draw a network diagram for the the following dependency relationship.

A and B control F; B and C control G.

Answer:

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Operations Research (MTH601)

Question: 48 (Marks: 3) Attempted Questions: 16 Total Questions: 52

Express the following Transportation problem(T.P) table into algebraic form with proper objective function and non-negative constraints

	D_1	D_2	D_3	Supply
O_1	c_{11}	c_{12}	c_{13}	5
O_2	c_{21}	c_{22}	c_{23}	3
O_3	c_{31}	c_{32}	c_{33}	10
Demand	4	6	8	

Answer:

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Operations Research (MTH601)

Question: 49 (Marks: 5) Attempted Questions: 16 Total Questions: 52

Check whether the given initial basic feasible solution is optimal or not.

0 5	2	1 1	6
2	1 5	5 2	7
2	4	3 7	7
5	5	10	

Answer:

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Operations Research (MTH601)

Question: 50 (Marks: 5) Attempted Questions: 16 Total Questions: 52

The cost of a new machine is Rs. 5000. The maintenance cost during the nth year is given by $M_n = 500\ (n - 1)$, where n = 1, 2, 3, 4, 5. If the rate per year is 0.05, after how many years will it be economical to replace the machine by a new one?

Answer:

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Operations Research (MTH601)

Question: 51 (Marks: 5) Attempted Questions: 16 Total Questions: 52

Wild West produces two types of cow boy hats. Type 1 hat requires twice as much labor time as a type 2. If all the variable labor time is dedicated to type 2 alone, the company can produce a total of 400 type 2 hats a day. The respective market limits for the two types are 150 and 200 hats per day. The profit is Rs. 8 per type 1 and hat Rs. 5 per type 2 hat.

Formulate the above linear programming problem.

(Note: solution is not required)

Answer:

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Operations Research (MTH601)

Question: 52 (Marks: 5) Attempted Questions: 16 Total Questions: 52

Write down the dual of the following Linear Programming problem;

$Min.Z = 3x_1 + 4x_2 + 6x_3$

Subject to:

$$x_1 + x_2 - x_4 \geq 10$$
$$-x_1 - x_2 + x_4 \leq -10$$
$$x_1 - x_3 = 0$$
$$x_2 + x_6 = 0$$
$$x_3 - x_7 = 0$$
$$x_1, x_3, x_4, x_5, x_6, x_7 \geq 0$$

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Answer:

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